

	Pre-thesis (Planning–Design)	Phase I Weeks 1–4	Phase II Weeks 5–8	Phase III Weeks 9–12	Phase IV Weeks 13–16
SDLC Stage Mapping	Planning – Design	Development	Development	Verification – Validation	Maintenance
Primary Focus	Requirements, Architecture, System & Smart Contract Design, Dataset & ML Pipeline	Core Contracts & Infrastructure Development	Lending Lifecycle, Oracle & ML Integration, Front-end & API	Testing, Security, Validation, Simulation, Documentation	Deployment Hardening, Monitoring, Optimization, Handover
Primary Deliverables	- SRS & Use Cases 4-tier Architecture Design - Smart Contract Spec - DB Schema & API Design - ML Pipeline Design - Project Plan & Risk Register	- Four-tier Core Contracts (WB, NB, LB, Wallet Auth) - Role & Access Control - Wallet Integration - Basic Frontend Shell - DB Schema & Migrations - Unit Tests (Core)	- Lending & Deposit Modules (SavingsVault, FixedDeposit, GroupLendingPool, IBLP, UpwardDeposit) - Oracle Integration - ML Scoring API (RF + IF) - Authority Brief (SHAP) - Frontend Flows (End-to-End)	- Integration & System Tests - Security Testing (Slither, Mythril) - Simulation (300 clients) - Formal Methods (Invariants) - Documentation (Tech + User) - Performance & Gas Analysis	- Testnet Deployment - Monitoring & Alerting - Bug Fixes & Optimization - Backup & Recovery - Final Report & Handover - Future Work Roadmap
Optional Agent (dashed)	—	—	—	Conversational Agent (MCP) – Prototype Development	Agent Integration & Evaluation
Key Milestone/Outcome	Design Complete (Spec Baseline)	Core Platform Operational	End-to-End Loan Flow with ML Scoring	Validated & Tested System (MVT Ready)	Deployed, Monitored & Handover Complete

